



10935 - A radiative transfer survey for interpreting Little Red Dot spectra

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INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Mr. Hanpu Liu (PI)	Princeton University
Prof. Eliot Quataert (CoI) (CoPI) (US Admin CoI)	Princeton University
Dr. Yan-fei Jiang (CoI)	Flatiron Institute
Prof. Jenny Emma Greene (CoI)	Princeton University
Yilun Ma (CoI)	Princeton University

OBSERVATIONS

ABSTRACT

Little Red Dots (LRDs) are a new, mysterious class of AGNs revealed by JWST. Their surprising abundance across $z \sim 4-8$ suggests a common phase of early black hole–galaxy coevolution, yet their spectral energy distributions (SEDs), especially the pronounced Balmer breaks in several objects, cannot be explained by standard AGN–galaxy composite models. Recent radiative transfer work points instead to a cool ($T_{\text{eff}} \sim 5000$ K), low-density ($\rho \sim 1e-14 - 1e-11$ g/cm³), optically thick gas medium that reproduces the distinct SEDs seen by JWST. Confirming and quantifying this previously unexplored regime would fundamentally revise our understanding of accretion physics and black hole growth at cosmic dawn.

We propose a systematic radiative transfer survey to map the spectral properties of such low-density atmospheres and enable interpretation of current and future JWST data. Using the TLUSTY radiative transfer code, we will compute a grid of high-resolution spectra across temperature, density, and metallicity and develop quantitative diagnostics of gas conditions, such as the Balmer break strength and Ca II triplet equivalent widths. The resulting public spectral library will let the community infer the physical parameters of LRDs analogously to stellar atmosphere fitting and guide future JWST observation to refine gas property measurements. In addition, we will assess the applicability of the Cloudy photoionization code for LRDs, which is widely used to infer optically thick gas from the Balmer breaks despite its design for optically thin conditions only. This program will provide the

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theoretical foundation needed to transform JWST's LRD discoveries into a coherent physical picture.

OBSERVING DESCRIPTION